



A review of sensing technologies for arsenic detection in drinking water

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Abstract

Water is essential resource vital for economic prosperity, well-being of ecosystems worldwide. Presence of arsenic, even in low concentrations, poses persistent threat due to its high toxicity, stability, its capacity to accumulate within food chains, affecting human health. To combat this issue, detection methods are striving to achieve lower limit of detection than World Health Organization standard of 10 parts per billion (133.3 nM), particularly focusing on toxic form, As^{3+} . Various techniques have been developed to detect, quantify arsenic in drinking water, including spectroscopic and electrochemical methods. Spectroscopic methods offer high sensitivity and selectivity but can be complex, costly to operate, while electrochemical methods, although simpler, cost-effective, may sacrifice some sensitivity, selectivity. Recent years have witnessed emergence of portable, field-deployable arsenic sensing devices, primarily based on electrochemical or optical principles, with potential to transform arsenic contamination monitoring. This review explores recent advancements in arsenic detection techniques in drinking water, highlighting substantial progress in development of highly sensitive and selective methods. It covers range of sensor technologies, such as electrochemical sensors, optical sensors, and nanomaterial-based sensors, all of which offer improved detection limits, accuracy. It discusses integration of emerging technologies like machine learning and miniaturized devices for real-time arsenic monitoring in water sources. Practicality and cost-effectiveness of these techniques are evaluated, emphasizing need for robust, field-deployable sensors to ensure access to safe drinking water in both developed and developing regions. This review aims to contribute to ongoing efforts to mitigate the arsenic contamination crisis and protect public health.

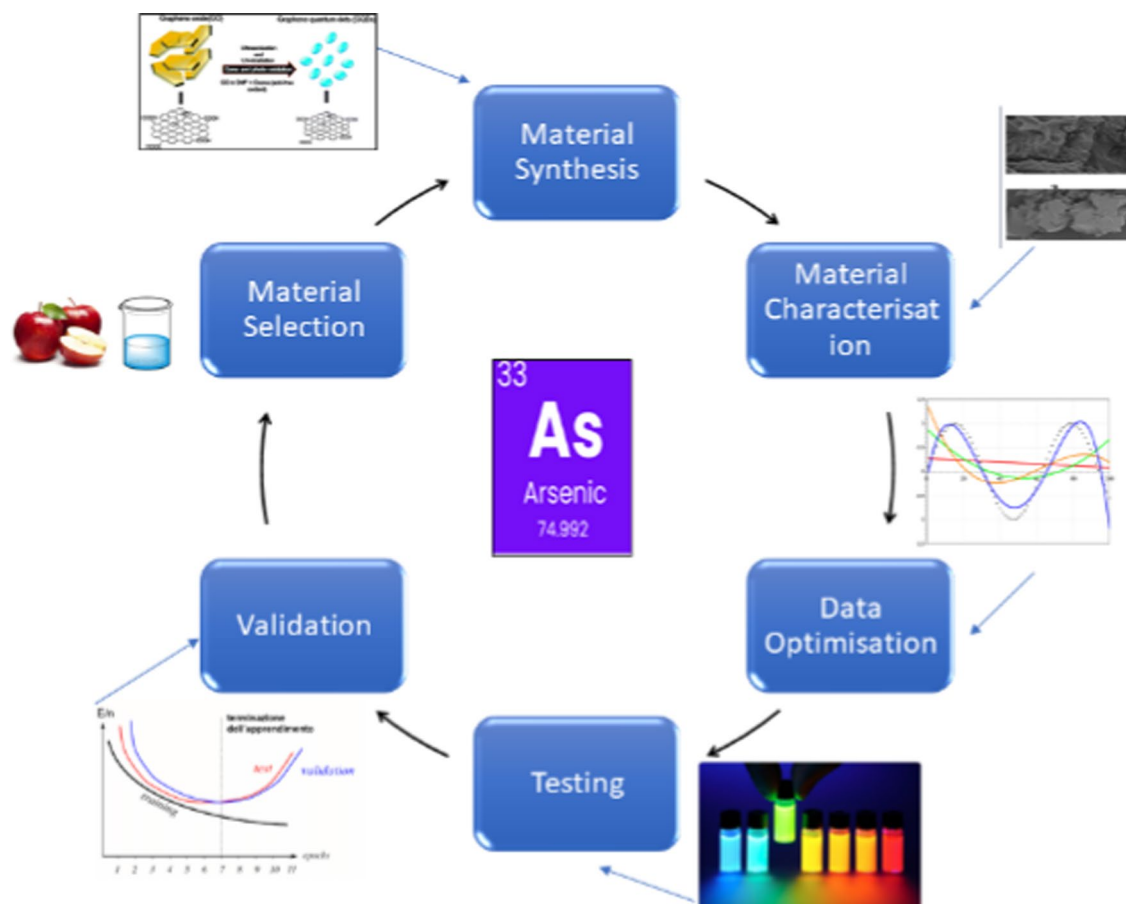
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Graphical abstract



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Introduction

Arsenic contamination in drinking water is a global health crisis, causing severe issues like cancer, skin ailments, cardiovascular disorders, and diabetes. This toxic element, found in the environment, is especially dangerous in its inorganic form, prevalent in acidic water (pH 2–11) (Aguirre et al. 2011). Chronic exposure, mainly through water and food, leads to serious health conditions. With increasing awareness, there is a growing demand for affordable arsenic detection methods in drinking water. New approaches using colorimetry, electrochemistry, and spectroscopy are being developed to address this need (Kim et al. 2019; Wang 2020; Chen et al. 2022).

Nanomaterials, with particles ranging from 1 to 100 nm, have unique properties making them useful in catalysis, environmental applications, sensors, and medicine. Their high surface-to-volume ratio, quantum size effect, and lattice contractions enhance their performance. In photocatalysis,

photon energy higher than the band gap energy generates electron–hole pairs, driving the reaction. Nanostructures are effective in sensing arsenic. Production techniques for nanocrystals include gas condensation, mechanical alloying, and chemical precipitation, while vapor deposition, sputtering, and electrodeposition form crystal layers. Sol–gel methods create clusters, and gas condensation and mechanical alloying produce large quantities of nanocrystals, enabling novel materials with new characteristics (Balusamy et al. 2020; Ahmad et al. 2019; Duran et al. 2016).

Nanoporous sensing materials like aerogels, zeolites, and core–shell structures exhibit novel properties and are synthesized through various methods (Bonyar et al. 2017). Factors influencing nanoscale particle size include geometrical configuration, attractive force, ionization energy, and various physical properties. Nanostructures differ from bulk materials due to surface and quantum confinement effects (Shen et al. 2019). Traditional solid-state physics and quantum chemistry are not fully applicable to nanomaterials, whose



properties depend on size. These properties include strength, diffusion, ductility, electrical resistivity, thermal expansion, luminescence, and magnetic characteristics. Nanomaterials differ from micromaterials due to increased surface area-to-volume ratio and quantum size effects, leading to higher surface energy, faster reaction rates (surface effect), and changes in magnetism, absorption, wavelength, and frequency (volume effect) (Cheng et al. 2019). Researchers have developed several analytical methods to detect arsenic in drinking water, aiming to improve sensitivity and detection limits (Devi et al. 2019). The most promising methods include electrochemical, surface sensing, spectroscopic, and colorimetric techniques, offering rapid, reliable, and highly sensitive arsenic detection to meet permissible limits (Li et al. 2019).

Nano-sensors have three parts such as material, recognition device and machinery for signal transduction. Novel optical, magnetic, or electrical features give rise to nano-sensors, which capitalize on the unique properties exhibited at the nanoscale (Fan et al. 2018). The method of signal transduction utilized to create a signal-following relationship with an analyte involves converting the presence or concentration of the analyte into a measurable signal, typically through mechanisms such as fluorescence, electrochemical reactions, or surface plasmon resonance (Zhang et al. 2021; Ahmad et al. 2019).

Researchers are focusing on the development of nano-sensors for high accuracy, selectivity, field applicability, economic efficiency, and promptness. Arsenic is tasteless and odorless, making it difficult to detect in drinking water (SC et al. 2020). Understanding the interrelationship between the arrangement of atoms and properties involves the characterization of materials at either atomic or nanometer scales. For this purpose, microscopic methods for detecting arsenic encompass various techniques, such as scanning electron microscopy (SEM), atomic force microscopy (AFM), transmission electron microscopy (TEM), and field ion microscopy (FIM). These methods are employed to visualize and analyze arsenic in samples without the need for extensive chemical processing or reactions. Some indirect methods are also used for their characterization such as Absorption spectroscopic techniques, X-ray diffraction of electrons or neutrons Rutherford backscattering (RBS) method, Raman spectroscopy, Auger electron spectroscopy (AES), Photoluminescence and Photoluminescence excitation techniques (Abdul et al. 2015; SC et al. 2020).

A colorimetry method requiring minimal manual actuation is effective for detecting As(III) ions, with a sensitivity of 1 g/l in the range of 5–20 g/l. The rGO/Fe₃O₄ NC enhances absorption due to the formation of mono and bidentate complexes and the superparamagnetic properties of magnetic nanomaterials at room temperature. Stripping voltammetry offers a consistent, cost-effective, and stable

sensor (Wei et al. 2020). The SERS method detects nanoparticles at the 1 ppb level in water (Shen et al. 2020). A colorimetric sensor using silver nanoparticles changes color upon reacting with arsenic, providing a simple and inexpensive detection method (Shrivastava et al. 2016).

A new electrochemical sensor based on gold nanoparticle-modified electrodes utilized their approach using cyclic voltammetry, which offered high sensitivity and selectivity for the detection of arsenic. The operation of this sensor and its performance parameters such as detection limit, response time was improved through the use of gold nanoparticles which makes it different from conventional sensor (Gregory et al. 2015). A quantum dot-based fluorescence sensor was developed, utilizing the fluorescence quenching effect caused by arsenic ions. This sensor was notable for its high sensitivity and low detection limit (Wang et al. 2017). A handheld paper-based sensor was developed that utilizes a smartphone, combining the simplicity of paper-based sensors with the high analytical performance of smartphones. The features of this method include portability, ease of use, and real-time data collection for field operations (Celikbas et al. 2018).

A biosensor was designed using genetically engineered bacteria that produce a detectable signal in the presence of arsenic. This biosensor is highly specific, allowing the detection of different forms of arsenic with high sensitivity and selectivity (Zhao et al. 2019). An optical sensor based on the SERS technique developed for the detection of glucose in diabetic patients. Their method employed silver-coated gold nanoparticles to amplify the Raman signal of arsenic (Xu et al. 2018). The novelty of this sensor lies in its ultra-low detection limits and its capacity to provide a molecular profile of arsenic species, which greatly aids in understanding the nature of arsenic pollution (Zhang 2020).

A nanocomposite sensor was designed using MOFs and CNTs, employing electrochemical impedance spectroscopy (EIS) for arsenic detection (Wu and Ju 2015). The inclusion of graphene oxide, with its high surface area and good conductivity, enhanced the sensor's performance in terms of sensitivity and detection limit (Sung et al. 2021).

A dual-mode electrochemical-optical sensor was developed, utilizing two modes to achieve better and more accurate results for arsenic concentrations. This sensor was particularly notable for its ability to detect a wide range of arsenic concentrations with a high degree of accuracy (Wang et al. 2019; Ji et al. 2022). A proposed wearable sensor for real-time arsenic detection uses flexible electronics and ion-selective electrodes for continuous monitoring, crucial for preventing exposure. This innovation, alongside advancements in nanomaterials, biological components, and analytical tools, enhances arsenic detection's sensitivity, selectivity, portability, and user-friendliness, addressing a critical social issue in drinking water safety (Choi et al. 2023).



The research on emerging sensing technologies for arsenic detection in drinking water, spanning from 2015 to 2024, highlights significant contributions from North America, Europe, and Asia. These regions, facing critical needs due to widespread groundwater contamination, have driven the development of advanced nanomaterial-based sensors, portable detection systems, and innovative analytical techniques. These advancements have substantially improved sensitivity, selectivity, and cost-effectiveness in arsenic detection. This review synthesizes findings from diverse geographical locations, providing a comprehensive overview of the technological progress made in addressing this vital environmental and public health issue.

This review comprehensively examines current analytical techniques for arsenic detection, comparing sensitivity, detection limits, and concentration ranges. It covers various methods, their performance, and capabilities, while discussing the challenges and future research directions for rapid, on-site water analysis and environmental monitoring. The review informs water and environmental analysis professionals about advanced techniques' critical role in protecting global water resources. It highlights efficient, cost-effective methods suitable for field use, addressing obstacles and proposed solutions. Innovations in nanotechnology, biosensors, microfluidics, portable devices, AI, and miniaturized technologies are showcased, emphasizing their potential to revolutionize arsenic detection for safe drinking water.

Review of conventional detection techniques

Several conventional analytical methods used for the determination of arsenic in drinking water are described in this review. It includes colorimetric, electrochemical, and spectroscopic techniques, as well as their advantages and limitations. A review of over many articles revealed that the most common methods for detecting arsenic in drinking water are colorimetric methods.

Top-down and bottom-up approaches are two distinct methods for quantifying arsenic concentration in a sample. The top-down approach involves first measuring the total concentration of arsenic in the sample, and then using a chemical separation method to isolate the arsenic. However, the top-down approach is not as sensitive as the bottom-up approach, and it can be difficult to separate the arsenic from other elements in the sample. Bottom-Up Approach is a more recent method for measuring arsenic. It is more sensitive than the top-down approach, and it can be used to measure arsenic in a wider range of matrices. However, it is more complex and costly than the top-down approach (Wang et al. 2019).

Arsenic is a highly toxic metalloid that can pose a severe hazard to human health, including an increased risk of cancer. Thus, it is crucial to be able to detect arsenic in water, food, and other environmental samples to safeguard public health. Conventional methods such as AAS and ICP-MS are very sensitive and selective for arsenic detection. However, they can be expensive and require specialized equipment. Emerging methods such as electrochemical biosensors, chemical sensors, and biosensors are also very sensitive and selective for arsenic detection, and they have the potential to be more portable and less expensive than conventional methods as shown in Fig. 1. Further research and development is needed to address challenges such as complexity, cost, and interference (Xu 2023).

Chromatographic methods

Chromatographic techniques are crucial for the accurate detection and quantification of arsenic in drinking water. These methods separate chemical compounds based on their interactions with a stationary phase and a mobile phase. High-Performance Liquid Chromatography (HPLC), Gas Chromatography (GC), and Ion Chromatography (IC) are commonly used for arsenic detection, each offering distinct advantages. They ensure drinking water safety by precisely identifying and measuring arsenic levels. Size-Exclusion Chromatography (SEC) is also used, contributing to the analysis of arsenic through its unique separation capabilities (Liu et al. 2015). Chromatographic methods are frequently combined with detection techniques like ICP-MS and AAS to enhance sensitivity

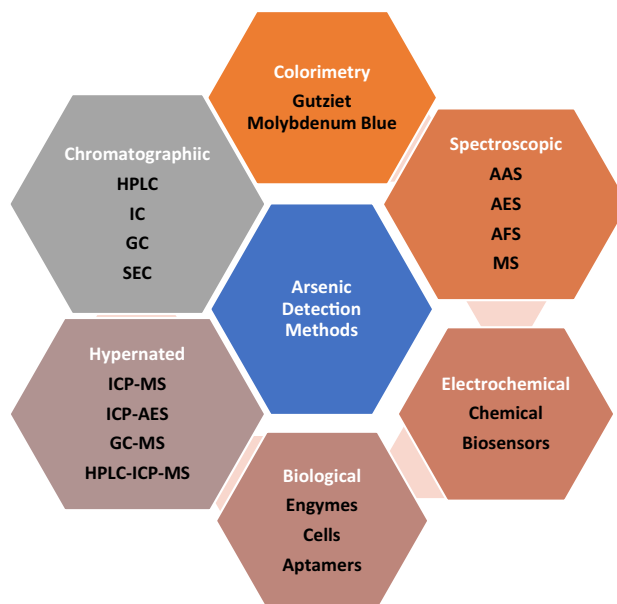


Fig. 1 Arsenic Detection Methods



and selectivity. ICP-MS offers high sensitivity for detecting arsenic, while AAS, though less sensitive, provides specificity for individual arsenic species (Cheng et al. 2018) as given in Table 1.

Chromatographic methods can be used to separate and quantify different forms of arsenic. These techniques have significant utility in identifying potential sources of arsenic contamination and in evaluating the risks associated with exposure to various arsenic forms. Furthermore, their versatility allows for the analysis of diverse sample types, encompassing drinking water, food, soil, and biological tissues (Zhao et al. 2018).

Biological methods

Arsenic detection in drinking water using biological methods relies on living organisms or their components to identify and measure arsenic levels. Biological methods can be more sensitive and selective than traditional chemical methods, and they are often less expensive and easier to use. Some of the most promising biological methods for arsenic detection includes Enzymes, Cells and Aptamers (Kumar and Arya 2022).

Aptamers are short nucleic acid sequences that can specifically bind to arsenic. Aptamers can be used to detect arsenic by monitoring their binding to the toxin. This can be done using a variety of methods, such as fluorescence spectroscopy and surface plasmon resonance (Zhang et al. 2022a, b). Biological methods for arsenic detection are a promising new approach for monitoring drinking water quality. Biological methods are sensitive, selective, and often less expensive and easier to use than traditional chemical methods. However, more research is needed to address the challenges of specificity, stability, and complexity before biological methods can be widely adopted as shown in Table 2.

This is because biological components can be produced on a large-scale using fermentation and other biotechnology techniques. Biological methods are also often easier to use than traditional chemical methods. This is because biological methods can be designed in a variety of formats, such as biosensor strips, kits, and assays (Gregory et al. 2015). Biological methods for arsenic detection face challenges such as lack of specificity, potential for false positives due to interactions with other compounds, and instability of biological components. Additionally, these methods can be complex and time-consuming, involving multiple steps like cell culture and enzyme assays. Despite these issues, they hold promise for revolutionizing arsenic detection in environmental and biological samples (Ahmed et al. 2020; Huang et al. 2023).

Table 1 Summary of Chromatographic Methods for Arsenic Detection

Chromatographic Method	Working Principle	Advantages	Limitations	Applications for Arsenic Detection	References
High-performance liquid chromatography (HPLC)	Separates compounds by pumping a liquid mobile phase through a solid stationary phase at high pressure, which separates the components of a sample based on their interactions with the stationary phase	High sensitivity and selectivity, wide range of applications	Can be complex and time-consuming, requires specialized equipment	Widely used for arsenic detection in water, food, and biological samples	Liu et al. (2015)
Ion chromatography (IC)	Uses a mobile phase to separate ionic compounds based on their charge and size	High sensitivity and selectivity for ionic arsenic species	Limited range of applications, requires specialized equipment	Can be used for arsenic speciation in water samples	Cheng et al. (2018)
Gas chromatography (GC)	Uses a carrier gas to separate volatile compounds based on their boiling points and interactions with the stationary phase	High sensitivity and selectivity for volatile arsenic species	Limited range of applications, requires specialized equipment	Can be used for arsenic speciation in air and water samples	Cheng et al. (2018)
Size-exclusion chromatography (SEC)	Classifies molecules according to their size	Used to separate arsenic species based on their molecular weight	Somewhat sensitive and selective, but not ideal for all applications	Capable of identifying the specific form of arsenic present in water and biological samples	Zhao et al. (2018)



Table 2 Summary of Biological Methods for Arsenic Detection

Method	Working principle	Advantages	Limitations	Applications	References
Enzymes	Enzymes catalyse a reaction that produces a measurable product	High sensitivity and specificity, relatively inexpensive	Limited to specific arsenic species, may be susceptible to interference from other compounds	Water quality monitoring, food safety testing, environmental monitoring	Kumar and Arya (2022)
Cells	Cells are monitored for their response to arsenic exposure	High sensitivity and specificity, can detect different arsenic species	May be time-consuming and complex to implement, may be susceptible to interference from other factors	Environmental monitoring, biomedical research, drug discovery	Zhang et al. (2022a; b)
Aptamers	Aptamers bind to arsenic specifically	High sensitivity and specificity, can be developed to detect different arsenic species, relatively stable under different conditions	May be expensive to produce, may be susceptible to interference from other compounds	Water quality monitoring, food safety testing, environmental monitoring, biomedical research	Huang et al. (2023)

Atomic absorption spectroscopy (AAS)

It is an extremely sensitive and specific technique for detecting arsenic. It is based on the principle that arsenic atoms absorb light at specific wavelengths. AAS is widely used in water quality monitoring and other applications. Advantages of this spectroscopic methods for Arsenic detection are high sensitivity and selectivity, ability to detect and quantify different forms of arsenic as well as versatility and applicability to a wide variety of samples. Limitations of this method can be expensive and require specialized equipment and methods can be complex and time-consuming to perform (Irunde et al. 2022).

Inductively coupled plasma mass spectrometry (ICP-MS)

ICP-MS is a highly sensitive and selective method for arsenic detection, based on measuring arsenic ions' mass-to-charge ratio. The process involves introducing a sample into a plasma source for vaporization and ionization, then using a mass spectrometer to segregate ions by their mass-to-charge ratio and detect arsenic by signal intensity. ICP-MS's versatility extends to detecting various elements and compounds, often enhanced by coupling with techniques like liquid chromatography (LC) or gas chromatography (GC) for separating and identifying arsenic forms in complex samples. This synergy makes ICP-MS crucial for arsenic analysis and widely applicable in various fields (Patel et al. 2017).

Colorimetry

Colorimetry is a straightforward and cost-effective method for arsenic detection, compared to techniques like AAS or ICP-MS. It works by converting arsenic into a colored compound that is detectable with a colorimeter. This technique is particularly useful in resource-limited settings due to its affordability and simplicity (Senthamizhan and Uyar 2020). Colorimetry measures substance concentration by comparing light absorption of the sample to that of a standard solution. The Gutzeit molybdenum blue method is a specific colorimetric approach where arsine gas reacts with a molybdenum blue solution, forming a blue complex that is quantified using a colorimeter, demonstrating colorimetry's practicality in arsenic analysis (Sanghavi 2022). The best method for arsenic detection depends on the specific needs of the application. For high sensitivity and selectivity, AAS and ICP-MS are ideal. For portability and low cost, emerging methods like electrochemical biosensors, chemical sensors, and biosensors are more suitable as given in Table 3.



Table 3 Comparison of Different Conventional Methods

Method	Working Principle	Advantages	Limitations	Similarities	Applications	References
Atomic Absorption Spectroscopy (AAS)	Based on arsenic atoms' light absorption at specific wavelengths	High sensitivity and selectivity, can detect and quantify different forms of arsenic, versatile	Expensive, requires specialized equipment, can be time-consuming	AAS and ICP-MS are both spectroscopic techniques	Water quality monitoring and more	Irunde et al. (2022)
(ICP-MS)	Arsenic ions are vaporized, ionized into a plasma, then detected based on their mass-to-charge ratio	Very sensitive and selective, can detect a wide variety of elements and compounds	Requires specialized equipment, often used in conjunction with chromatography	ICP-MS is a spectroscopic technique	Water quality monitoring and more	Patel et al. (2017)
Colorimetry	Arsenic is converted to a colored compound and quantified using a colorimeter	Simpler and less expensive, suitable for resource-limited settings	Less sensitive and selective than spectroscopic methods	All methods can detect and quantify arsenic	Particularly used in developing countries, Gutzeit molybdenum blue method for arsine gas detection	Sanghavi (2022)

Review of emerging detection techniques

Detecting arsenic in drinking water faces major challenges, including the need for rapid, cost-effective methods suitable for routine monitoring, especially in developing countries. Many traditional laboratory-based methods are too expensive and time-consuming for frequent use. Furthermore, selective detection is complicated by the presence of other elements like iron and manganese sulfide, which can interfere with colorimetric and electrochemical methods. There is growing demand for affordable, specific arsenic detection techniques. Recent advancements have introduced new methods utilizing colorimetric, electrochemical, and spectroscopic technologies to address these challenges and improve the efficiency and accuracy of arsenic monitoring (Rehman et al. 2016). Promising new methods for detecting arsenic in drinking water include:

Paper colorimetric sensors

These sensors produce a colored material based on the reaction of arsenic with a colorimetric reagent. The sensor changes color depending on the arsenic concentration in the sample. Low-cost and user-friendly paper-based colorimetric sensors can detect arsenic concentrations as low as 1 µg/L.

Electrochemical sensors

These sensors are based on the electrochemical properties of arsenic. Arsenic can be detected by measuring the voltage or voltage produced when a sample containing arsenic is placed in an electrochemical cell. Electrochemical sensors are sensitive and selective for arsenic and can be portable and cost-effective to manufacture (Siddiki et al. 2012).

Surface-enhanced Raman spectroscopy (SERS) sensors

These sensors are based on the interaction of light with arsenic-containing samples. Arsenic can be detected by measuring the SERS spectrum of the sample (Mulvihill et al. 2008). SERS sensors are highly sensitive and selective for arsenic and have the potential to be manufactured (Álvarez et al. 2010).

Electrochemical biosensors

Electrochemical biosensors detect arsenic by generating an electrical signal from the interaction of a biological molecule with a transducer surface. Enzymes specific to arsenic catalyze reactions that produce measurable outputs, such



as current or pH changes, which are directly related to the arsenic concentration in the sample (Zhao et al. 2022). Electrochemical biosensors have the potential to be more portable and less expensive than traditional methods for arsenic detection. They can also be made to be more selective for arsenic, which can reduce the risk of false positives (Umapathi et al. 2022).

Chemical sensors

Chemical sensors for arsenic detection typically use a chemical reaction that produces a measurable product, such as a change in color or fluorescence. The quantity of the product is directly linked to the arsenic concentration within the sample. Chemical sensors have the potential to be more portable and less expensive than traditional methods for arsenic detection. They can also be made to be more selective for arsenic, which can reduce the risk of false positives (Bu et al. 2015).

Biosensors

Biosensors for arsenic detection use antibodies or other biological molecules that specifically bind to arsenic, producing measurable signals like changes in electrical conductivity or optical properties. They offer advantages over traditional methods, being more portable and cost-effective (Zhang and Guo 2022). These biosensors also provide enhanced selectivity, reducing false positives. Emerging optical and quantum dot-based methods further improve portability, affordability, and ease of use compared to conventional techniques such as AAS and ICP-MS (Zhang et al. 2019).

Optical methods

Optical methods for arsenic detection involve light interaction with arsenic atoms or molecules, resulting in changes in light intensity, wavelength, or polarization. These changes help determine arsenic concentration. Two key techniques are used in optical sensors: Surface Plasmon Resonance (SPR) and fluorescence spectroscopy. SPR detects arsenic by measuring changes in the refractive index of a gold surface upon arsenic binding, indicating its concentration (Samanta and Shunmugam 2021). Fluorescence spectroscopy, on the other hand, involves binding arsenic to a fluorescent dye, which emits light when excited by a different wavelength. The brightness of the emitted light correlates with arsenic concentration in the sample (Elcin and Öktem 2020; Gebremedhin 2024).

Nanomaterials based methods

Nanomaterials have unique properties that make them promising for the detection of arsenic in drinking water. Nanomaterials can be engineered to have high surface areas, which allows them to adsorb more arsenic from the water (Babu and Doble 2018). Nanomaterials can also be designed to have specific optical or electronic properties that can be used to detect arsenic. Nanomaterials have potential to revolutionize the way that arsenic is detected in drinking water. Nanomaterials can be used to develop arsenic detection devices that are more sensitive, selective, and portable than traditional methods. Additional research and development are essential to tackle the obstacles associated with the utilization of nanomaterials in arsenic detection, including issues related to expenses, potential toxicity, and dispersion. In the Fig. 2 various properties of nanomaterials is given.

Metal nanoparticles, such as gold and silver, detect arsenic using Surface Plasmon Resonance (SPR), which measures changes in the refractive index of the nanoparticle surface caused by arsenic binding (Samanta and Shunmugam 2021). Carbon nanotubes are employed in fluorescence spectroscopy for arsenic detection. Here, a fluorescent dye coated onto carbon nanotubes emits light when bound by arsenic (Boakye et al. 2019). The intensity of the emitted light correlates with arsenic concentration in the sample (Ge et al. 2019; Gebremedhin 2024).

Quantum dot-based methods

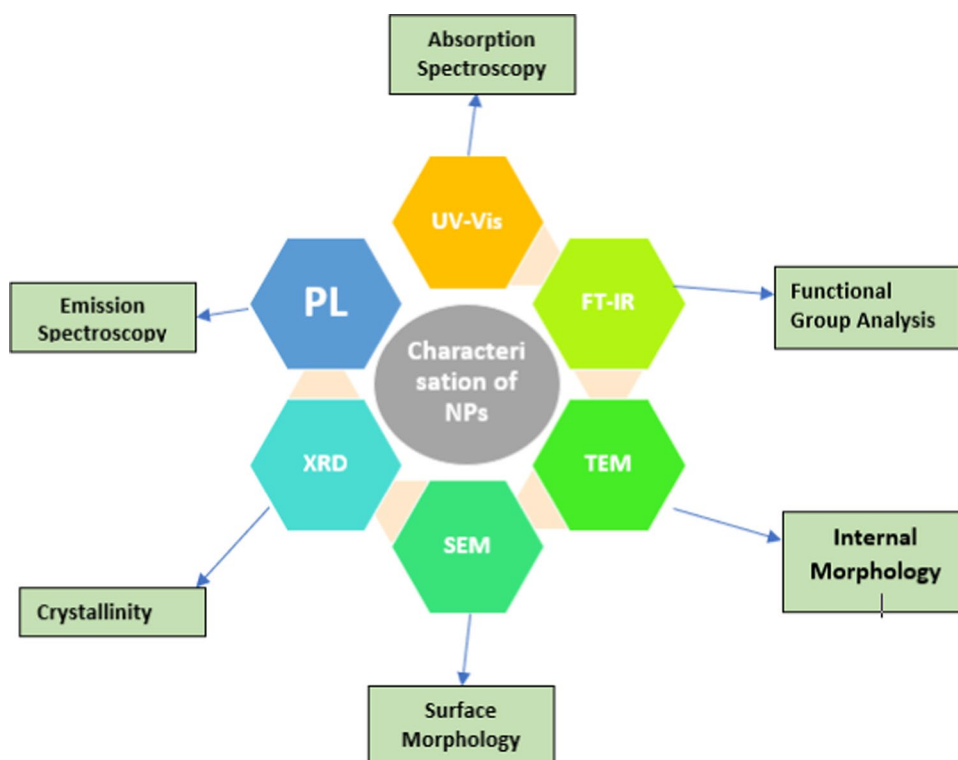
Quantum dots, semiconductor nanoparticles with unique optical properties, can detect arsenic by binding to arsenic-specific antibodies. This binding causes the quantum dots to emit light, with the intensity of the emission correlating to arsenic concentration. While quantum dot-based methods are promising for their sensitivity, selectivity, and cost-effectiveness, they are still under development and face challenges. Issues include sensitivity to sample interferents and complexities in production (Rahimi et al. 2021; Bhat et al. 2023). Emerging arsenic detection methods, including optical and quantum dot-based techniques, offer potential but require overcoming hurdles before widespread adoption. The review highlights various arsenic detection methods based on selectivity, sensitivity, limit of detection, and cost, approaches for accurate and economical arsenic monitoring as discussed in Table 4.

Case studies

Several case studies showcase diverse methods for arsenic detection in drinking water, highlighting their efficiency and practicality. Colorimetry, a cost-effective approach, employs



Fig. 2 Properties of Nanomaterials



color changes to detect arsenic, as seen in Gutzeit molybdenum blue's blue complex formation. Electrochemical methods leverage arsenic's electroactive nature for precise quantification, with studies demonstrating their effectiveness in real-time monitoring. Raman spectroscopy presents a non-invasive and label-free solution, offering rapid arsenic identification through molecular vibrations, as demonstrated in field-based applications. Innovative approaches like carbon dot (C-dot) synthesis integrate nanotechnology to enhance sensitivity, with successful case studies illustrating the improved detection limits and selectivity, addressing arsenic contamination challenges in drinking water (Ferrari and Basko 2013). These case studies collectively reflect the adaptability and progress in arsenic detection techniques, catering to various needs and settings.

Colorimetry method

Colorimetric techniques have been preferred for portable arsenic monitoring because of their simplicity and the method of detection. Colorimetry and spectrophotometry are related techniques which both depend on the interaction of molecules with light, specifically, the absorption and emission characteristics. Colorimetry can be done by naked eye or through digital camera; however, spectrophotometers provide greater accuracy because they quantify transmittance or reflectance at several wavelength intervals. Nonetheless, spectrophotometers are relatively costly than colorimeters

in the market. In the case of field applications, the use of portable digital detection instruments like cameras, UV–Vis spectrometers or even smart phones increases the practicality of colorimetry. Microfluidic systems that are tailored specifically for the colorimetric tests enhance the flexibility, usability, and reliability of colorimetric tests (Patey et al. 2008). Colorimetric methods, especially for the determination of arsenic (III), are characterized by the simplicity of the method, portability, semi-automation, and low actuation. These methods show the detection limit of 1 µg/l and are efficient within the linear scale of 5–20 µg/l for As (III) ions. The temperature effect on these devices is also taken into account to make sure that the device will work accurately to the temperature required (Bhatia et al. 2014).

One of the major strengths of colorimetric methods is the creation of portable, easy to use instruments that are fully automatic and capable of data storage, especially for countries that have limited resources. However, some limitations are present, for example, the Gutzeit reaction and Hach test kit do not meet WHO criteria because of sensitivity problems and rapid fading of color within a short period of time. Economical aspect shows that the cost of manufacturing of 20 prototypes and 100 microfluidic cartridges is 700 USD while the market price for the same is approximately 1 USD (Idris et al. 2017) 0.5 USD per test, which is cheaper and more reasonable compared to other available tests. However, these economic factors are not the priorities, and the focus is still given to the



Table 4 Comparison of Different Emerging Methods

Method	LOD ($\mu\text{g/l}$)	Sample Size	Time	Advantages	Limitations	Applications	Future Directions	References
Colorimetric Sensors	0.1–10	Small	Rapid (minutes)	Simplicity, cost-effectiveness	Limited sensitivity, specificity	Resource-Limited Settings	Enhancement of sensitivity and selectivity, automation, integration with digital technologies	Sanghavi (2022)
Electrochemical Sensors	0.001–1	Small	Rapid (minutes)	High sensitivity, selectivity	Electrode fouling, stability issues	Environmental Monitoring, Water Quality	Miniaturization, enhanced materials, field deployment	Siddiki et al. (2012)
Surface-enhanced Raman Spectroscopy (SERS)	0.001–0.1	Small	Moderate (minutes)	High specificity, label-free detection	Limited sensitivity for low concentrations	Trace Analysis, Materials Characterization	Signal enhancement, nanomaterial advances	Jiang et al. (2017)
Electrochemical Biosensors	0.001–0.1	Small	Rapid (minutes)	Specificity, biocompatibility	Biofouling, need for complex bio-elements	Health Monitoring, Environmental Analysis	Miniaturization, biorecognition elements	Huang et al. (2023)
Chemical Sensors	0.01–10	Varies	Rapid (minutes)	Diversity of analytes, real-time data	Limited to specific chemical compounds	Industrial Safety, Environmental Monitoring	Enhanced selectivity, data analysis techniques	Bu et al. (2015)
Biosensors	0.01–5	Small	Rapid (minutes)	Biocompatibility, specificity	Shelf-life of biological components	Medical Diagnostics, Environmental Monitoring	Integration with microfluidics, improved stability	Zhang et al. (2022a; b)
Optical Methods (e.g., SPR)	0.001–0.1	Small	Rapid (minutes)	Real-time analysis, label-free detection	Limited LOD for some analytes	Biomedical, Environmental Analysis	Enhancing LOD, integration with other techniques	Rajendran et al. (2021)
Fluorescence Spectroscopy	0.001–0.1	Small	Moderate (minutes)	High sensitivity, specificity	Photobleaching, limited to fluorescent compounds	Biological and Environmental Applications	Advanced fluorophores, signal processing	Gebremedhin (2024)
Nanomaterials-based Methods	0.001–1	Small	Rapid (minutes)	Enhanced sensitivity and selectivity	Synthesis challenges, cost	Various Analytical Fields	Improved nanomaterial design and production	Ge et al. (2019)
Quantum Dot-based Methods	0.01–0.5	Small	Rapid (minutes)	High brightness, tunable properties	Toxicity concerns, potential environmental impact	Biomedical, Environmental Analysis	Developing non-toxic alternatives	Rahimi et al. (2021)



simplicity of the process. In general, colorimetric methods, especially with the help of microfluidic systems, can be considered as a convenient and effective approach to arsenic detection in portable format maintaining sensitivity, accuracy, and cost-efficiency (Cui et al. 2015).

Working principle

In colorimetry method, the process is initiated by Hach Arsenic test kit and it is inserted into microfluidic cartridge, instructions are displayed by LCD and lighting by LED. Monitoring of microcontroller and sensor readings give measurement results and output is obtained in form of temperature rise, sensitivity and kinetic response (Oliveira et al. 2015).

The Hach® EZ Arsenic compact Test Kit (8.07 × 5.71 × 9.06 inches) is a portable kit used in the field, that uses a microfluidic cartridge employing the Gutzeit reaction. During this reaction, hydrogen is converted to sulfate to remain non-interference in a neutral field, and inorganic arsenic is reduced to arsine gas (AsH₃) with the help of powdered zinc and sulfamic acid. Arsine gas when brought in contact with the mercury bromide (HgBr₂) on test paper produces arsenic/mercury halogenides such as AsH₂HgBr (Nguyen et al. 2018). This method depends on the rate of discoloration which is directly proportional to the concentration of arsenic, but it cannot differentiate between As (III) and As (V). It works well in acidic solutions but can be affected at a pH of less than 5 and by sulfide at 15 ppb, antimony at 250 ppb, and selenium at 1 ppm (Lin et al. 2017).

The cartridge of the kit is an upper frame of polyurethane, a lower flame retardant frame of 1.4 mm thickness and a mid-frame of micro-fluidic elastic polymer film containing reagents. The device operates automatically and the instructions are shown on the LCD and LEDs. First, a red arrow signals to turn the valve, and then a green arrow signals that it is properly aligned. An air of 50 ml is injected to the solution to spread the Zn powder before adding 50 ml of distilled water (Abalos et al. 2012). The measurement area is illuminated by four WS2812B RGB LEDs, and an Omni Vision OV9655 CMOS sensor takes the picture (Xie et al. 2012). An image is received and processed by a microcontroller, and the relative blue constituent is compared with a reference field image. The Hach kit with the label HgBr₂ acts as the standard and shows the concentration of arsenic using red and green LED when the sample is introduced.

The water sample is measured with the help of CNY70 optical sensor and the device is charged with 750 mAh, 3.7 V Li-ion battery with the 12 V DC step-up voltage (Gupta 2018). The result of the Colorimetry method has found and shown in Fig. 3.

Outputs

In Colorimetry method, the time-concentration graph indicates that there is a constant concentration for some time as indicated in the Fig. 3a. The time-temperature graph also indicates that there is constant temperature for some time this is the average of 6–13 measurements taken at 50 °C as indicated in the Fig. 3b. In colorimetric method it is observed that, Various Arsenic (III) concentration with time shows maximum deviation at 60 min at ambient temperature of 22 °C in Fig. 3c. Different Arsenic (III) concentration with time depicts the superimposed deviation at 25 min at the ambient temperature of 50 °C as shown in Fig. 3d (Bon-yar et al. 2017).

Parameters

In this method, mainly two parameters are given as **sensitivity** and **elevated water temperature**. The approach mentioned assesses the purity of water according to the sensitivity and the high temperature of the water. The sensitivity here refers to the ability of the technique to distinguish between drinking and non-drinking water through measurement of the levels of arsenic. For the WHO regulatory limits are 10 µg/L (ppb) for industrialized countries while for developing countries is 10 µg/L (ppb) and 50 µg/L (ppb). The method has a sensitivity of about 1 µg/L with a detection limit of around 3 µg/L that complies with IUPAC standards (Zhang et al. 2021; Devi et al. 2017). Elevated water temperature impacts the kinetics and measurement accuracy of arsenic concentrations. Stable arsenic levels are reached in 60 min at 22 °C and 25 min at 50 °C. However, the points of inflexion used to approximate arsenic concentrations are significantly different from standard values. Temperature is primarily a factor in experimental settings rather than construction. To minimize response time, temperatures from 25 to 80 °C are analyzed, though sample preparation should not exceed 50 °C to prevent vapor formation. This method is suited for users with less technical expertise, offering lower precision (Kempegowda et al. 2014; Devi et al. 2017).



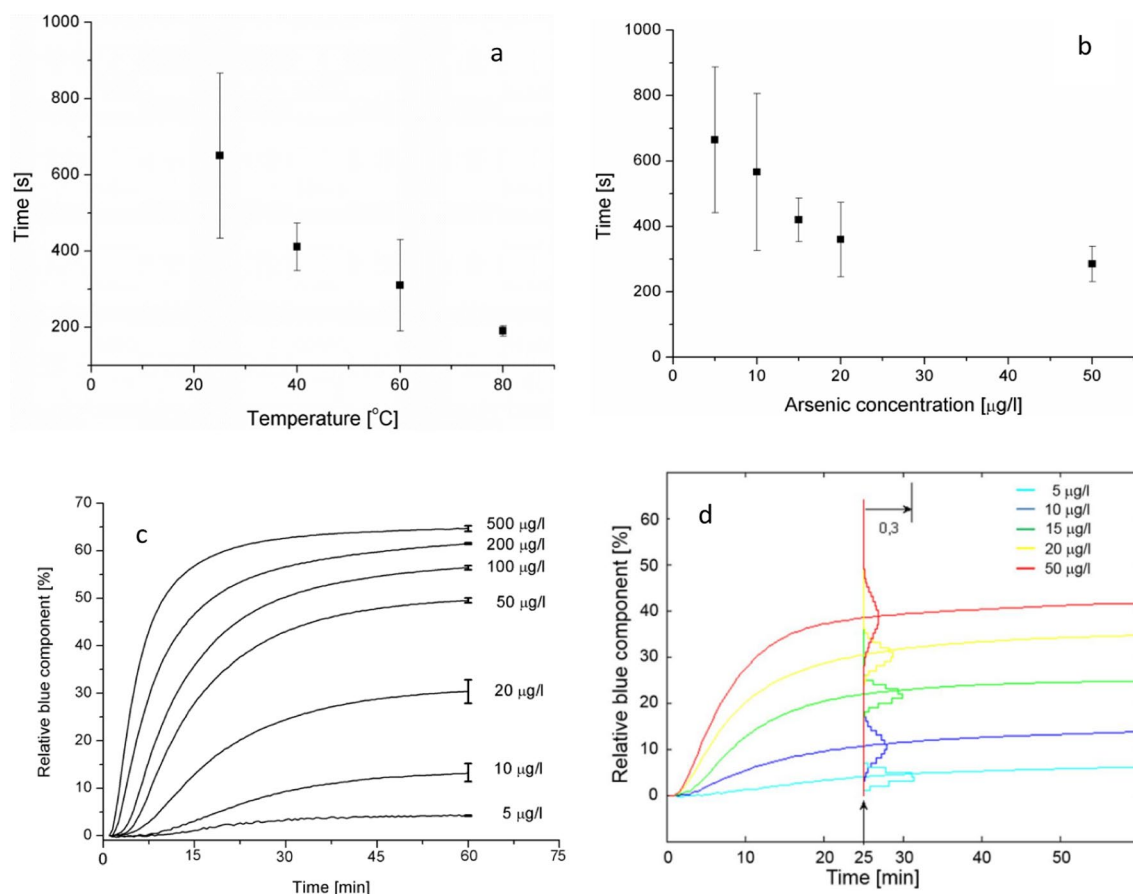


Fig. 3 Results of Colorimetry method. **a** The appearance of the kinetic curves inflexion points as a function of sample Temperature observed at 20 g/l Arsenic Concentration As (III). With the exception of 80 °C, when only 3 measurements were taken, the average and standard deviation of 6–15 measurements were taken (Bonyar et al. 2017). **b** The inflexion point of the kinetic curves, when considering an arsenic concentration of 20 g/l (III), is influenced by the sample temperature. It's worth noting that there were only three measurements conducted at 80 °C, resulting in an average and standard deviation

range of 6–15 (Bonyar et al. 2017). **c** Kinetic curves of various Arsenic (As (III)) concentrations measured at laboratory ambient temperature (22 °C). The error bars at 60 minutes show the results' min-max variation based on 3–5 measurements (Bonyar et al. 2017). **d** The kinetic curves represent experiments conducted at a temperature of 50 °C, involving various arsenic concentrations (As (III)) ranging from 5 to 50 grams per liter. The distributions overlaid at the 25-min mark are calculated based on measurements from 5 to 13 (Bonyar et al. 2017).

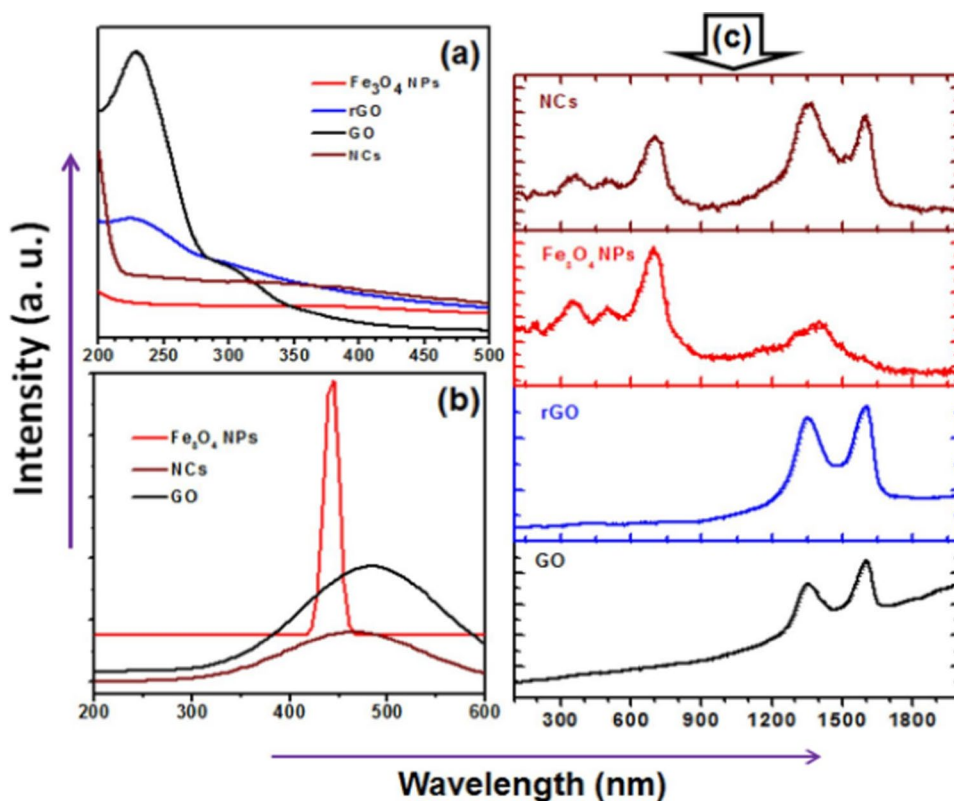
Electrochemical method

Electrochemical methods are very efficient for arsenic detection in drinking water, of which stripping voltammetry is more suitable for field application because it is sensitive, cost effective, and provides a quick survey with precise computation (Devi et al. 2017). However, these methods have their drawbacks, for example, non-recyclable materials, the methods' range of detection, and high grounding currents. Au and its nanostructures

(NSs) are popular because of As-Au bimetallic alloy formation, although this is expensive. Scientists are working on carbon-based composites containing metal nanoparticles for arsenic detection due to their opto-electrical characteristics and high surface area (Hignett et al. 2004). Graphene-Pt NCs with CV and SWASV exhibit a linear relationship in the 10–100 nM range and LOD of 1 nM. Concentrations of 1 nM for arsenic (III) were used. Other prospective materials include rGO-lead dioxide (rGO-PbO₂) NCs, electro-reduced graphene oxide (ErGO)-Au



Fig. 4 Optical characterizations of GO, rGO, Fe_3O_4 NPs, and NCs as synthesized: UV–vis, PL, and Raman spectroscopy are the three types of spectroscopy (Zandsalimi et al. 2011)



NPs, and metal NPs with graphene oxide (Ag-GO) with the LOD of 0.24 nM. Moreover, the rGO/ Fe_3O_4 nanocomposites have a very high adsorption capacity for arsenic and can be used for both sensing and purification. These new materials and approaches are assessed based on their efficiency, replicability, and stability (Li et al. 2015; Zhao et al. 2019).

Working principle

In electro-chemical method first, we select a material then material synthesis and processing is performed followed by instrumentation and different characterization techniques and nanocomposites synthesis and processing is done then nanoelectrodes preparation is done. In this process, initially we select material in which all chemicals utilized were of analytical grade and employed without additional purification. Graphite flakes, H_2SO_4 , 97%, H_3PO_4 , 85%, HCL, 35%, NH_4OH , 25% H_2O_2 , 30%, NaOH, KH_2PO_4 , FeCl_3 , KMnO_4 , 99.9%, As_2O_3 , $\text{K}_3\text{Fe}(\text{CN})_6$, $\text{K}_4\text{Fe}(\text{CN})_6 \cdot 3\text{H}_2\text{O}$, KCl are used for different purpose.

For synthesis and sensing, Lead (II) chloride, Lead Nitrate, Zinc Nitrate hexahydrate, Chromium Chloride, Deionized (DI) Water (18 M Ω), and pure N_2 (99.

Experiments were carried out with a CHI140A electro-chemical workstation comprising of three glassy carbon electrodes, a platinum counter electrode and a saturated calomel reference electrode (Zandsalimi et al. 2011). Characterization involved the use of SEM and TEM for the morphological and structural studies. SEM images showed uniformly dispersed Fe_3O_4 NPs synthesized via a wet chemical pathway, appearing as bright spots on wrinkled rGO sheets at a 5:1 feed ratio, meaning that the loading density is high, but the polymer chains do not aggregate. The TEM images also supported the uniform dispersion of Fe_3O_4 NPs in rGO sheets with the NP size decreased from 5 to 3 nm due to the influence of GO on size and morphology (Zhou et al. 2018). UV and PL analysis were also performed as well as Raman spectroscopy using Renishaw 2000inVia spectrometer with 514 nm excitation wavelength (Wei et al. 2016).

The GO of Asbury's graphite in an acidic environment was followed by barium chloride addition to ensure that there was no acid in the final product. Fe_3O_4 NPs were deposited on self-reduced rGO in an in-situ manner directly in the reaction solution. Dispersing GO (25 mg) in de-ionized water (10 mL) and adding 25% NH_4OH to adjust the pH to ~10, FeSO_4 (125 mg) was added and the mixture was stirred at room temperature for 24 h. The final black



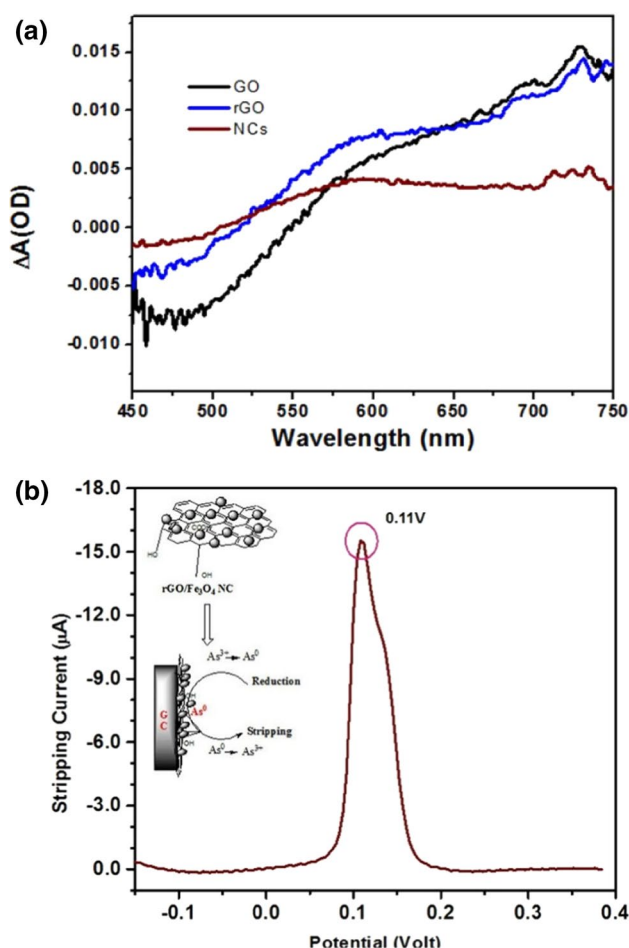


Fig. 5 NCs/GCE electrodes' SWASV analytical characteristic toward Arsenite and rGO, Fe₃O₄ NPs, and NCs were measured using UFS Transient Absorption (TA) Spectroscopy (Zandsalimi et al. 2011). **a** The NCs/GCE electrodes' SWASV analytical characteristic toward Arsenite (Zandsalimi et al. 2011). **b** rGO, Fe₃O₄ NPs, and NCs were measured using UFS Transient Absorption (TA) Spectroscopy (Zandsalimi et al. 2011).

suspension was prepared with rGO/Fe₃O₄ NCs, which was centrifuged at 4000 rpm and washed with DI water and then dried at 60 °C for 4 h. The glassy carbon electrode (GCE) was then polished with 1 μm and 0.05 μm alumina slurry and washed with isopropanol and dried under a stream of N₂ gas. For GCE modification, 5 μL of rGO/Fe₃O₄ (1 mg/mL) solution was then applied onto the GCE surface and allowed to air dry. Electrochemical studies were carried out using CV and EIS with a 5 mM Ferri/Ferro electrolyte in 0.1 M KCl (Vinothkannan et al. 2015; Zowada and Foudaz 2019). For As (III) detection, optimized parameters were frequency:

40 Hz, amplitude: 0.025 V, and incremental capacity: 0.004 V in PBS (pH=5) under stirring condition. SW-ASV included the step of As (III) reduction at −0.2 V for 180 s and measuring in the range of −0.2 V to 0.4 V. Selectivity studies challenged the nanocomposites/GCE with As (III) in the presence of Co, Cd, and Zn at 10 times the concentration of As (III) (Martinez et al. 2018).

Outputs

For the optical characterization of GO nanocomposites, this involves the study of Intensity-Wavelength graphs with Photoluminescence, Raman Spectroscopy and UV-Visible Spectroscopy. The SWASV Analytical characteristics of NCs/GCE for Arsenate are determined under specific optical conditions. Current–Potential curves are shown in cyclic voltammetry for nanocomposites. UFS Transient Absorption (TA) spectroscopy is used to check out nanocomposites. The swasv reaction of NCS/GCE electrodes towards arsenate at different concentrations is observed.

Figure 4 illustrates GO, rGO, Fe₃O₄ NPs and NCs as synthesized by means of UV–Visible, PL and Raman spectroscopies. Figure 5a reveals SWASV analytical features of NCs/GCE electrodes for arsenite while Fig. 5b shows UFS Transient absorption (TA) measurements rGO, Fe₃O₄ NPs and NCs.

Scientists use cool tools to check out graphene oxide (GO), reduced graphene oxide (rGO), and rGO/Fe₃O₄ nanocomposites. They look at these materials with UV–visible, photoluminescence, and Raman spectroscopy. The Raman spectra show how messy the crystal structures are in GO, rGO, Fe₃O₄ nanoparticles (NPs), and nano-composites (NCs). They show with a 514 nm laser to get these results (Zandsalimi et al. 2011). Transient Absorption (TA) Spectroscopy lets them peek at super-fast electronic stuff happening in femtoseconds. This gives them the scoop on how carriers move around. Scientists use cyclic voltammetry in Fe(CN)₆^{3−}/KCl electrolyte to study how Fe₃O₄ NPs, GO, rGO, and nano-composite grey carbon electrodes behave electrochemically. The Randles–Sevcik equation helps figure out diffusion coefficients for reversible and diffusion-controlled reactions.

The slopes and intercepts may be anticipated through the use of Eqs. 1 and 2:

$$ipc = 33.9 \times 10^{-6} \times v^{1/2} + 0.978 \quad (1)$$



Table 5 Suggested sensor for detecting As(III) is compared to other sensors

Sensors	Sensitivity ($\mu\text{A/ppb}$)	LOD (ppb)	References
Au NPs/GCE	0.24	0.0096	Álvarez et al. (2010)
Au-C Composite Electrode	0.133	0.375	Álvarez et al. (2010)
MnO _x /Au NPs	0.193	0.0057	Álvarez et al. (2010)
MWCNT/Au	0.236	–	Álvarez et al. (2010)
RTIL-Fe ₃ O ₄ /GCE	4.91	8×10^{-4}	Das et al. (2014)
Au-Fe ₃ O ₄ /GCE	0.113	0.83	Álvarez et al. (2010)
CoO _x /GCE	0.00148	0.825	Das et al. (2014)
rGO-Fe ₃ O ₄ /GCE	0.281	0.12	Das et al. (2014)

$$ipc = -42.47 \times 10^{-6} \times v^{1/2} + 0.90874 \quad (2)$$

The Randles–Sevcik equation is employed to calculate the diffusion coefficient for reversible and diffusion-controlled responses (Zandsalimi et al. 2011). Electrochemical Impedance Spectroscopy gives a Nyquist plot that shows electrode surface properties. Semicircles in the plot point to charge transfer resistance, while straight lines mean diffusion is happening. A special formula calculates the electron transfer constant for the NCs/GC electrode. The cost of the price switch resistance (RCT) relies upon the dielectric and insulating capabilities on the electrode/electrolyte interface (Zhao et al. 2017). The electron switches consistent (ke) for the NCs/GC electrode are calculated with the aid of using

$$ke = RT \ n2F2RCTA \quad (3)$$

SWASV shows better sensitivity than cyclic voltammetry for As(III) detection with NCs/GO electrodes (Zandsalimi et al. 2011). NCs' electrocatalytic properties cause this great performance, which rGO and Fe₃O₄ NPs boost even more. Scientists twisted things like deposition time and potential to make arsenite detection work better. They found the best setup for square wave anodic stripping voltammetry. pH levels really shake up iron oxides' stability and arsenite sorption. Higher pH cuts down peak current and messes with electrostatic interactions (Ali et al. 2014; Cui et al. 2012).

Suggested sensor for detecting As(III) is compared to other sensors

Several electrodes based on sensitivity and the limit of detection to As(III) are discussed, among which RTIL-Fe₃O₄/

GCE was found to have the highest sensitivity with respect to the limit of detection. Therefore, it is the most efficient electrode in the detection of As(III). On the other hand, CoO_x/GCE had the lowest sensitivity with the highest limit of detection (Das et al. 2014). Table 5 compares the proposed As(III) detection sensor with a few other existing sensors in terms of performance evaluation for its sensitivity, specificity, and overall efficiency of the new sensor. In the comparison between sensors, it can be observed that the RTIL-Fe₃O₄/GC sensor EtB is the most sensitive, while the Au-Fe₃O₄/GCE had the highest limit of detection.

SERS method

The low concentration of Arsenic in Drinking Water is determined by HPLC–MS, (ICP–MS), atomic fluorescence spectroscopy (AFS) methods. This method can detect the low concentration of As but cannot differentiate between oxidation state. To get vast knowledge for toxicology, remedy and controlling measures surface-active substrate method is surmounted these are many few methods which detection limit is below WHO acceptable limit and Environment Protection Agency(EPA) (Jiang et al. 2017). The Arsenic problem usually more exist in the western part of India. This method is expensive and time-consuming by this spectroscopy method As can be detected in three different ways biosensor, SERS sensor and electrochemical sensor. SERS method is more reactive to pesticide (Han et al. 2017).

SERS technology is used for environmental review and its controlling measures. Some of the inorganic pollutants determined by this method are Arsenate (AsO₄^{3−}), chromate (CrO₄^{2−}), cyanide (CN[−]), perchlorate (ClO₄), thiocyanate (SCN[−]). Nowadays gold nanoparticles or colloidal silver are the most widely applicable SERS substrates but non-reproducible. Most SERS substrates are metallic surface and transition metals and few of the non-metallic substrates are available. Some examples of metallic SERS substrates are copper, silver, gold and examples of transition metals are nickel, copper, platinum, and palladium. Some of the experiments are performed on semiconductors such as zinc sulfide, lead sulfide, copper oxide, TiO₂, zinc oxide. In this concept surface defects increase the probability of electron transfer which enhances the SERS phenomenon (Ahmad et al. 2018).

Many silica oxides, copper oxide, manganese hydroxide of gold and silver are reported such as Au–SiO₂, Au–TiO₂, Ag–CuO, Au–MnOOH, Ag–MnOOH are investigated. The most accurate result for Arsenic detection below 10^{−10} M



level is determined by using Ag–FeOOH SERS substrate. So, metal to metal oxide and hydroxide nanostructures are available effective and economical substrates for this method, for these electromagnetic and chemical properties, are responsible (Yadav et al. 2016).

SERS improvement depends on electric field intensification near the metal surface when resonance occurs between oscillating conduction electrons with an incident ray in phase. The dimensions, configuration and closeness of nanoparticles are affected by the frequency and magnitude of the localized surface photon which directly affects the degree of Raman enhancement. The observations such as dimension and configuration can be done by SEM and TEM techniques. Electromagnetic coupling helps to increase the excitement of surface photons. Silver is one of the strongest photon-responsive elements and gold is also used because of its chemical strength and affinity for many laser wavelengths. Langmuir Blodgett (LB) assembly was used for making silver nanostructures. PVP (polyvinyl pyrrolidone) is made of silver air and Water fixed and synthesis is done by polyol process. Uniformity of dimension and the configuration are obtained using the LB process. The pressure range for silver nanoparticles is 14 MnCm². SERS enhancement factor (EF) is a data-based measurement of Raman amplification. $EF = (I_{\text{surface}})/(I_{\text{solution}})$ BN surf ace/N solution (Yang et al. 2020).

The quantity of molecules examined within a standard solution and on the substrate is denoted as N-Solution and N Surface, while the signal intensities in both normal and SERS (Surface-Enhanced Raman Scattering) modes are represented as I solution and I surface. To ascertain the number of molecules interacting with the surface, the total surface area of each nanocrystal was divided by the Vander Waals dimensions of the thiol head group, assuming a close-packed monolayer formation of benzenethiol (Hao and Meng 2017).

Experimental section

The literature reveals the methodology. Nanoparticle preparation is done by the polyol method for silver which are quasi spherical with a diameter ranging in nano-meter. In this method PVP is mixed with ethylene glycol heated in an oil bath it is converted to light yellow from colorless. Another mixture of ethylene glycol with silver nitrate is heated for up to 4 h. Then cooled down to room temperature (Litter et al. 2019; Bundschuh et al. 2019).

In this method (SERS) substrate is used made of stable FeOOH-impregnated nanoparticles. Ag surface dendrite

in nature is acquired by galvanic reaction under ice-cold conditions (Li et al. 2012). One of the limitations of this method is not preferable for field detection because the time required for sample making is long. One of the issues of this method is the development of a substrate for the determination of a single element and sensitivity increment and reproducibility (Gowen et al. 2012).

Carbon dot synthesis method

High-performance liquid chromatography-mass spectrometry, inductively coupled plasma mass spectrometry, and atomic fluorescence spectroscopy usually make detection for low concentration levels of arsenic in drinking water. Most of these techniques detect low levels of arsenic but fail to make distinctions between oxidation states. Most surface-active substrate methods provide solutions for detailed toxicological analyses and remediation strategies. However, most of these techniques either have high costs or are highly time-consuming, and the accomplishment of detection limits below WHO and EPA standard is still challenging (Jiang et al. 2017). Surface-Enhanced Raman Spectroscopy SERS is a highly reactive method applied for the detection of various inorganic pollutants such as arsenate, chromate, cyanide, perchlorate, and thiocyanate. SERS makes use of metallic nanoparticles like gold and silver or transition metals like nickel and platinum as substrates for the process. This will then enhance the Raman signal through localized surface plasmon resonance of these substrates. Silver is definitely brilliant as a photon responsive element. Gold, on the other hand, is valued for its chemical stability and affinity towards the wavelength of lasers (Han et al. 2017; Ahmad et al. 2018).

Recent improvements have tested nanostructures like gold–silica, gold–titanium dioxide, silver–copper oxide, and silver–manganese oxyhydroxide for arsenic detection. Of these, the sensing substrates of silver-iron oxyhydroxide have reported the best results for the detection of arsenic below 10^{−10} M. These metal-oxide and metal-hydroxide nanostructures have emerged as efficient and cost-effective due to their electromagnetic and chemical properties combine (Yadav et al. 2016). SERS relies on electromagnetic field intensification near the metal surface, where the dimensions, configurations, and proximity of the nanoparticles dictate the Raman signal. Therefore, techniques like SEM and TEM are performed to observe those characteristics. Usually, the SERS EF is defined



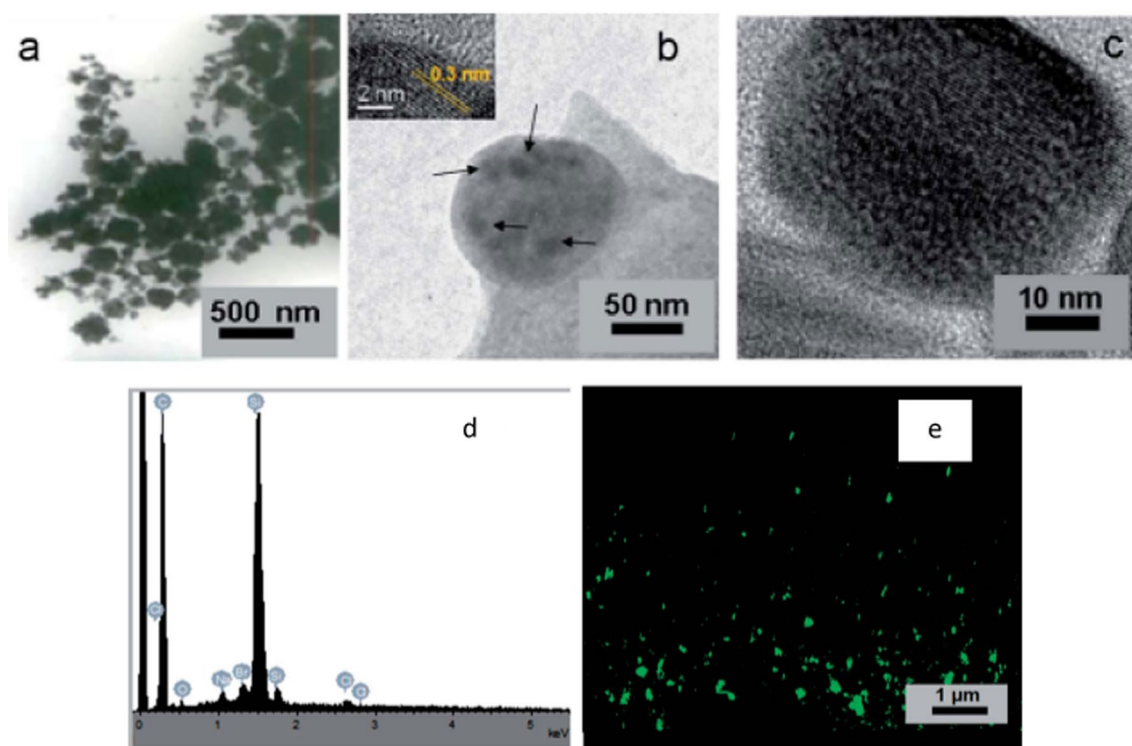


Fig. 6 TEM and EDAX image of the as-prepared meso-SiO₂/C-dot complex. **a** TEM image of the as-prepared meso-SiO₂/C-dot complex, **(b)** enlarged TEM image of selective dots embedded in meso-SiO₂ particles, with highlighted spots being carbon-dots adhered to meso-SiO₂ particles **C-** and the inset showing a HRTEM snapshot

of a representative meso-SiO₂/C-dot particle with a fringe width of 0.3 **(c)** HRTEM of porous meso-SiO₂/C-dots with C-dots adsorbed on them **(d)** The complex's EDAX, which depicts the major elements composition, displays the complex' fluorescent picture (Pandey et al. 2014).

to quantify Raman signal enhancement, taken as a ratio of the signal intensities and the number of molecules at the surface versus in solution (Yang et al. 2020; Hao and Meng 2017).

Working principle

The RHC preparation process initiates with 5 g of rice husk powder, which was washed with deionized water, dried, and ground to a mesh size of 100. Put the powder in a furnace under an atmosphere of nitrogen at 750 °C for two hours and obtain rice husk ash. This RHA is accordingly treated with NaOH (99.9%) at 900 °C in an N₂ atmosphere. After the treatment process, rice husk carbon is washed with DI water, filtered under vacuum, and dried at 90 °C for 12 h. In the second part, a preparation of Carbon Quantum Dots (CQDs) is undertaken. 50 mg RHC is mixed with 10 ml H₂SO₄ (98) and 3 ml DI water. After that, the mixture will be ultrasonicated for 5 h. Add 20 ml HNO₃ (70%) and further

ultrasonicate for another 10 h. The resulted black solution will be vacuum filtered through a 0.22 μm membrane. The filtered sample was treated with 30 mL of DI water, and the pH adjusted to 8 by use of 1 M NaOH solution. Heating this mixture in a 100 mL Teflon autoclave at 200 °C for 10 h makes it functional. Thereafter, it is subjected to vacuum filtration through a 0.22 μm membrane to obtain rice husk-derived CQDs. In order to store the CQDs, they are preserved in a 100 mL ethanol solution. A 5 ml aliquot of this solution was added to 45 ml of DI water and then ultrasonicated for 5 h. Finally, centrifugation at 10,000 rpm for 30 min was performed to collect the small particles from the upper part of the vessel.

Outputs

Characterization is done for quantitative and qualitative analysis. As shown in Fig. 6 TEM image of carbon dots produced from rice husk and their EDAXs. The TEM image



Table 6 Comparative Review of Arsenic Detection Methods (Welch et al. 2013; Cui et al. 2015; Jiang et al. 2017; Spain and Venkatanarayanan 2014)

Parameters	Colorimetry	Electrochemical Method	Raman Spectroscopy	Carbon Dot Synthesis Method
Materials	Water	Graphite flakes, HCl, H ₃ PO ₄ , NH ₄ OH, KCl, KMnO ₄ , NaOH, H ₂ SO ₄ , lead nitrate, zinc nitrate, nickel chloride, cobalt chloride, chromium chloride, DI Water	SERS Ag-FeOOH substrate, gold nanoparticles, colloidal silver	Rice husk (coffee beans, ashes, waste), orange juice, eggshell membrane, CA, L-cysteine, dextrin, lactose, HCl, glucose, H ₃ PO ₄ , BSA, urea, citric acid, amine
Instrumentation	Test kit, microfluidic cartridge, microcontroller, LCD, LED, feedback device, calibration unit, power supply system	Three electrodes (working and reference electrode), spectroscopy stripping voltammetry	Langmuir Blodgett assembly, grating, interferometer, CCD, PMT, monochromatic source, UV, visible, NIR	Microwave digestion system for C-dot synthesis or conventional method
Effective Parameters	Effect of elevated water sample temperature, concentration, time relative component percentage, pH	High sensitivity, selectivity, field application, cost effectiveness, accurate results, pH, deposition time, deposition potential, stripping current	Raman wavelength, concentration, particle shape, peak area, substrate absorbing capacity, frequency, photon intensity, pressure, surface area, doping concentration	Fluorescence, wavelength, function group, reaction time, reaction rate
Limitations	Cannot detect below WHO guideline, temperature is limited, concentration is limited, accuracy is less	Issues related to reproducibility, robustness, detection range, sensitivity, electrode fragility, toxicity concerns, high background currents, challenging operational conditions	SERS controlled process, expensive, time-consuming sample formation, not preferable for field detection	None specified
Properties	Portable, semi-automated, minimum actuation	High sensitivity, selectivity, cost effectiveness, short analysis time	Electromagnetic and chemical characteristics, rapid response times, exceptional sensitivity, reliability, non-destructive, swift, simple, non-invasive	High quantum yields, chemical stability, fine-tunable surface chemistry, short reaction time, fast, simple, reproducible
Sensitivity	1 µg/L	0.00148–4.91 µA/ppb	10 ⁻¹⁰ M	High sensitivity
Limit of Detection	0.12 ppb	0.0008–0.825 ppb	2.1 ppb	Effective at very low concentrations
Deviation	16%	8%	3%	Not specified
Range	10–50 µg/L	50–200 µg/L	Not specified	Effective in various ranges
Medium	Acidic	Acidic-Basic (2–11)	Not specified	Not specified
Economic Factor	Not Very Costly	Medium Range	Costlier than Other Methods	Economical and efficient
References	Welch et al. (2013)	Cui et al. (2015)	Jiang et al. (2017)	Ahmad et al. (2018)



details the morphology and distribution of carbon dots, showing size, shape, and uniformity. The EDAX images give elemental composition, thus evidencing the existence of key elements that will be present in the carbon dots and therefore ensure the correct synthesis from rice husk. TEM images detailing the structure of the meso-SiO₂/C-dot Complex: Fig. 6a Overall view of the as-prepared complex Fig. 6b Close view in which selective carbon dots are embedded in the meso-SiO₂, while brightened spots represent carbon dots attached to the surface of the mesoSiO₂ particles. The inset of this figure represents a HRTEM snapshot of a representative meso-SiO₂/C-dot particle, in which the fringe width is 0.3 nm. Image Fig. 6c is representing the HRTEM view of porous meso-SiO₂/C-dots where carbon dots adsorbed on the porous structure of meso-SiO₂. Figure 6d EDAX analysis of the meso-SiO₂/C-dot complex. It shows the key elemental constitution. The analysis confirms major elements in the complex that can support the integrity of characteristics from a structural and compositional point of view for the synthesized material. In addition, EDAX results demonstrate fluorescent properties for the complex; therefore, since the synthesized material of the complex in this work is characterized by fluorescent properties revealed during EDAX studies, a potential application in a field such as fluorescence application by (Pandey et al. 2014).

Reviews of different methods

The summary of different methods based on their material, instrumentation, parameters, properties, concentration range, sensitivity and limit of detection. Based on sensor characteristics can be divided into two systems, first one is static which includes accuracy, distortion, threshold, non-linearity, hysteresis, Minimum-Signal-Detectable, non-linearity, Selectivity/Specificity, Sensitivity, Threshold and the second one is dynamic error response, noise, drift, instability, repeatability, step response etc. discussed in Table 6.

The colorimetry method entails the use of a Hach-kit, microcontroller, lighting system, and power supply system to detect water. It measures the water temperature, concentration and pH and therefore can be described as portable and semi-automated with little actuation. Nevertheless, its concentration and accuracy range is limited, as reviewed in Table 6 (Welch et al. 2013). The electrochemical method utilizes graphite flake and different chemicals; it applies stripping voltammetry to describe current–voltage characteristics. It measures pH, deposition time, deposition potential

and stripping current, however, it has some limitations such as; non-reproducibility of the material used, brittleness of the electrode, high background currents, yet it has high accuracy, good electrocatalytic properties and stability (Cui et al. 2015). This method, particularly with solid electrolytes, is safe and realistic in the identification of arsenic in drinking water. As a result of being used solid electrolytes with electrochemical sensors, they are a very safe and practical option to detect the presence of arsenic in drinking water, since they possess the characteristics of durability and selectivity (Spain and Venkatanarayanan 2014). The sensory system is prepared with materials such as graphene oxide and gold microelectrodes, which are amino group-modified to improve sensitivity and anti-interference properties (Arinicheva et al. 2020). One innovative method and approach for this task has been to make use of hybrid metal nanoparticles, gold and platinum in particular, in order to increase the detection limit and reliability of such sensors. (Such sensors are the catalyst to efficient monitoring and compliance with arsenic contamination limits in water bodies, thus providing a viable solution to environmental and health safety (Kate et al. 2018; Tufail et al. 2022). Raman spectroscopy uses gold nanoparticles and colloidal silver; NIR, grating and interferometer techniques are used. It measures Raman wavelength, concentration of the solution, particle shape, photon intensity, and doping concentration. Although this method is costly and requires a lot of time, it is characterized by good electromagnetic and chemical performance, low response time, high sensitivity, reliability, as well as non-destructive nature. The carbon dot synthesis method uses raw materials such as rice husk, synthesized through microwave digestion, and coffee beans. This method involves the study of fluorescence, wavelength, functional groups, reaction time, and rate to obtain carbon dots with high quantum yields, chemical stability and surface chemistry. Thus, the presented carbon dots have short reaction times, high response, sensitivity, reliability, non-destructive nature, high speed, easiness and repeatability (Jiang et al. 2017).

Three different methods are compared based on sensitivity, limit of detection, deviation, concentration range, medium and cost. Colorimetry method has lowest sensitivity, lowest limit of detection, works in acidic medium and cheap method whereas Raman spectroscopy has highest sensitivity, highest limit of detection and costly method (Cui et al. 2015; Jiang et al. 2017).



Challenges

Detection of arsenic in drinking water is important because arsenic exposure poses serious health risks. There are many challenges to arsenic detection in drinking water, and research is underway to address these challenges. Detecting arsenic in groundwater presents several challenges. The World Health Organization (WHO) has set a maximum allowable level (MCL) of 10 µg/L for arsenic in drinking water, but in some regions, arsenic concentrations can reach hundreds or thousands of µg/L, making detection difficult with traditional methods. Interference from ions such as phosphate, silicate, and sulfide complicate detection, requiring selective method. Accurate measurement at WHO MCL levels necessitates sophisticated, expensive equipment like ICP-MS and GF-AAS, which are scarce in developing countries due to cost and the need for skilled technicians. The cost and complexity of some detection methods limit their accessibility in rural areas. Complex matrices in natural water sources also affect accuracy, necessitating proper sample preparation (Khan et al. 2016). Limited laboratory access in remote areas highlights the need for simple, cost-effective field methods (Crecente-Campo et al. 2017). Long-term monitoring is crucial to ensure ongoing water safety, requiring methods adapted for continuous use (Bundschuh et al. 2012). Compliance with varying regional regulations is essential, necessitating methods that meet these standards (Foster and Maher 2019). Additionally, public awareness and education are vital for communities to understand and manage arsenic contamination in their water sources (Wong et al. 2015). Addressing these challenges involves developing innovative, accessible, and effective arsenic detection technologies. In addition to these challenges, there are specific challenges related to the detection of arsenic in drinking water in developing countries. For example, many rural people in developing countries rely on groundwater from wells and hand pumps. In addition, many developing countries lack trained sanitation workers and equipment for arsenic removal and treatment. Despite the challenges, there is a great need for accurate and inexpensive methods for arsenic detection in developing countries (Sardar et al. 2022).

Factors affecting sensitivity

Sensitivity is influenced by three primary factors. Chemical factors, including concentration, substrate-absorbing capacity, particle shape, and solvent binding properties,

impact sensitivity. Dimensional factors, such as response time, structural dimensions, intensity, area, and wavelength, also affect sensitivity. Furthermore, properties like thermal, optical, magnetic, electrical, and electro-chemical properties play a role in sensitivity. Detection of arsenic in drinking water depends on various sensing technologies, and several factors can affect the efficiency of these sensing methods. Arsenic detection in water faces several challenges. Coexisting ions like sulfide, phosphates, and silicates can interfere, necessitating high selectivity (Ahamad et al. 2018). Complex sample matrices require careful preparation (Foster and Maher 2019). Low concentration detection demands improved sensitivity (Smedley and Kinniburgh 2002). Ensuring accuracy and precision is crucial for reliable results. Cost and availability impact the adoption of sensing technology (Bundschuh et al. 2012). Portable equipment enhances field applicability (Khan et al. 2016). Long-term stability is vital for continuous monitoring. Sensor characteristics, environmental factors, and proper sample preparation also significantly affect detection accuracy (Li et al. 2018).

Conclusion

Water is an essential resource crucial for human life, economic development, and ecosystem health. Groundwater and surface water serve as vital sources of drinking water for millions globally. Contamination by toxic substances like arsenic poses a severe threat to human health. This review provides an overview of arsenic detection methods, focusing on As(III) and As(V) species in water, and explores techniques such as chromatography, spectroscopy, colorimetry, biology-based methods, electroanalysis, and combined approaches. It highlights current challenges and future research directions, emphasizing the need for improved analytical performance, rapid analysis, and on-site testing technologies. While hybrid methods like ICP-MS, LC-MS, and HPLC/ICP-MS are promising, the future lies in innovative on-site approaches using biosensors, nanomaterials, and electroanalytical devices.

Novel materials and nanomaterials enhance arsenic analysis through electrochemical, optical, and chromatographic methods. Electrochemical sensors, based on redox potential and conductivity, are highly sensitive but can be affected by other ions. Optical sensors, utilizing absorption and luminescence, are sensitive but hindered by water turbidity. Chromatographic sensors, which separate arsenic from other components, are less sensitive but less susceptible to interference.

Advances in arsenic detection, including nano-sensors, electrochemical sensors, optical sensors, and smartphone-based solutions, offer significant benefits. Machine learning and data analytics enhance measurement accuracy,



promising improved arsenic analysis. Despite challenges like ion interference and the need for affordable, portable devices, ongoing innovation and a multidisciplinary approach are crucial. Addressing these issues is essential for safe drinking water access and reducing arsenic-related health risks, highlighting the pivotal role of advanced analytical methods.

Future study

The future of arsenic detection in drinking water is promising with advancements in sensitive, selective, and affordable technologies. Innovations include nanoparticle-based, optical, and paper-based sensors, and improvements in existing methods. Key prospects involve point-of-use detection devices, integrated detection and removal systems, and machine learning enhancements. Advanced sensor technologies, nanotechnology, and smartphone-based detection will be crucial, alongside miniaturization and data analytics for improved accuracy and efficiency. The focus is on developing low-cost solutions and multiparameter sensors compliant with regulations. Community engagement, research collaborations, and international cooperation will drive advancements, supported by innovative materials and low-cost remediation technologies. There is a significant demand for an AI-based mobile application using smartphone cameras to measure parameters via deep CNN algorithms. A fluorescence-based smart optical sensor is a crucial future need, and AI-driven predictive tools will help ensure access to safe, clean drinking water, establishing early warning systems for arsenic-free water access.

The newest green high-tech technologies that accommodate the environmental friendliness, the green sensors for detecting arsenic in drinking water exhibit superb detection capabilities, while incorporating the innovative nano-coating ensured decreased assay time. These sensors are environment-friendly and offer financial controls whose use is significant in cases of excessive pollution. Further advancement will allow the integration of the portable devices and microfluidic systems with these sensors to acquire real-time data which will greatly help in the provision of safe drinking water all over the world. Besides this, the sensors could specifically meet the primary objective of guaranteeing safe drinking water worldwide once they get optimized in terms of sensitivity and specificity.

Environmental significance

Arsenic poisoning from contaminated drinking water is a major environmental health threat, with arsenic(III)/arsenic(V) species posing the highest toxicological risk. The WHO and EPA have established standards, restrictions, and limits on the amount of arsenic that can be found in drinking water in an effort to prevent arsenic contamination and the toxicity that goes along with it. While the current laboratory-based techniques are adequate for analyzing arsenic, they are costly, time-consuming, and necessitate careful sample preparation. Emerging needs include portable approaches that are quick, affordable, and dependable, as well as on-site sensor-based solutions. This review provides a comprehensive exploration of numerous analytical approaches used to detect and quantify arsenic in water. It aims to increase awareness regarding the critical role these methods play in safeguarding the environment and global water resources.

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